MAC Protocol Optimization for Heterogeneous Wireless IoT Networks

Sambu Srinidhi

1 Introduction

1.1 Evolution of Heterogeneous Wireless IoT Networks

The advent of widespread Internet of Things (IoT) systems has led to the development of heterogeneous wireless communication systems, which are highly heterogeneous in terms of communication protocols, transmission powers, computational powers, and priority levels of the applications. Unlike homogeneous wireless sensor networks, which utilize uniform hardware and protocol stacks, heterogeneous IoT systems include Wireless Body Area Networks, smart city infrastructures, industrial automation systems, agricultural monitoring systems, and 5G-based gateways in a single integrated communication system. The combination of short-range low-power sensors and long-range high-bandwidth sensors in heterogeneous systems makes the channel conditions, traffic levels, and resources heterogeneous. Such heterogeneous systems pose considerable complexity in the Medium Access Control (MAC) layer of the network, where the efficient coordination of the shared resources is essential.

In the case of the carrier-sense multiple access (CSMA) protocol family, which is the most popular protocol family for wireless communication systems, fairness issues arise due to the heterogeneous environment caused by the unbalanced sensing thresholds and traffic loads. The stronger the transmission capability of the nodes or the better the channel condition of the nodes, the more the throughput will be biased towards those nodes. As a result, the chances of collision will also increase for the weaker nodes. Yu et al. proved that the performance of the CSMA-based mechanisms will degrade considerably in a heterogeneous contention environment [13].

As heterogeneous IoT networks continue to evolve, the integration of intelligent adaptation mechanisms has become increasingly necessary to address these challenges. Reinforcement learning-based MAC strategies dynamically adjust contention parameters to mitigate unfairness and optimize spectrum utilization under varying traffic conditions [13]. Additionally, energy-aware scheduling and multi-objective optimization approaches

attempt to balance delay, throughput, and power consumption across heterogeneous devices. The evolution from static, rule-based MAC configurations toward adaptive and learning-driven architectures reflects the growing complexity of IoT ecosystems. Consequently, modern heterogeneous wireless IoT networks demand MAC protocols that are not only scalable and energy-efficient but also capable of intelligent adaptation to fluctuating network states and diverse application requirements.

1.2 Motivation for MAC Optimization

Energy efficiency and collision avoidance are two important issues in the design of heterogeneous IoT networks, considering the limited energy resources of sensor nodes and the shared nature of wireless communication channels. Frequent collisions, retransmissions, and idle listening operations have a major impact on the performance of these networks, particularly in terms of increased energy consumption. Deterministic scheduling techniques, such as Time Division Multiple Access, eliminate the occurrence of contention-related collisions in the wireless communication channels. Hassan et al. introduced a TDMA-based MAC protocol for WBANs, focusing on minimizing idle listening and retransmission costs by employing priority-aware transmission slots [7]. The authors have shown the impact of structured superframes on stabilizing the reliability of WBANs and extending the lifetime of sensor nodes.

Nonetheless, it should be noted that deterministic scheduling may not cope well with highly dynamic heterogeneous traffic conditions. In the dense environment of the smart city, IoT devices are characterized by unpredictable and bursty heterogeneous traffic, and hence, the tuning of the MAC parameters may be required. In light of these challenges, machine learning-based approaches have been proposed as alternatives for intelligent medium access optimization. Ibrahim and Hafez showed the effectiveness of the adaptive configuration of the 802.11ah protocol's parameters in reducing the probability of collision under dense and heterogeneous traffic conditions [8]. These observations emphasize the significance of intelligent optimization mechanisms with the ability to achieve the tradeoff between reliability, flexibility, and energy efficiency in heterogeneous wireless IoT systems.

1.3 Research Objectives and Scope

The main purpose of this study is to offer a comprehensive and critical analysis of the strategies for optimizing the MAC protocol in a heterogeneous wireless IoT network. The main aim of this study is to critically evaluate how deterministic scheduling, reinforcement learning, machine learning, collision avoidance, clustering, and cross-layer optimization address the issues arising from heterogeneity. In addition, there is a major focus on the trade-offs between reliability and adaptability, and the computational and power-related issues of intelligent optimization techniques. By integrating the contributions of various

recent studies, this study attempts to highlight the positive and negative aspects, and the trends, in the design of the MAC protocol.

The scope of this analysis will be to comparatively evaluate various performance metrics, such as energy efficiency, throughput, packet delivery ratio, delay, fairness, and scalability. Deterministic TDMA-based mechanisms will be analyzed for their reliability, whereas reinforcement learning and distributed learning will be analyzed for their adaptability to dynamic environments [13, 4, 9]. Machine learning-based collision avoidance strategies will be evaluated for their efficacy in dense heterogeneous environments [8]. Additionally, cluster-based routing and multi-objective optimization will be evaluated for their efficacy in providing energy and delay trade-offs [6, 11]. By providing a comprehensive analysis, this study aims to provide a structured basis for developing hybrid multi-objective MAC protocols for next-generation heterogeneous IoT environments.

2 Background and Conceptual Foundations

2.1 Architectural Characteristics of Heterogeneous IoT Systems

In heterogeneous IoT environments, there is a combination of various wireless communication standards, hardware capabilities, and application requirements, all integrated into a single networking paradigm. Typically, heterogeneous IoT environments will be comprised of IEEE 802.15.4-based low-power wireless sensor nodes, IEEE 802.11ah-based wireless devices for long-range wireless IoT communication, Bluetooth Low Energy-based wireless peripherals, and cellular 4G/5G wireless gateway technology. The combination of all these diverse wireless technologies will result in a variety of transmission range, data rates, and power consumption. As a result, there will be a need for a heterogeneous IoT-based MAC protocol to address asymmetrical wireless communication links, diverse wireless traffic types, and heterogeneous wireless capabilities. In this regard, there will be a combination of high-bandwidth wireless gateway technology and ultra-low-power wireless sensor technology, resulting in channel access asymmetry.

Another aspect of MAC coordination in a heterogeneous network is the management of mobility and handoff. In a 5G-integrated IoT network, there are mobility-aware handoff mechanisms for dynamic switching of communication links to minimize power consumption [10]. These changes in the network topology require the MAC protocol to be adaptable to changes in topology. Finally, there are issues of coexistence, resulting from interference between different technologies that operate in the same unlicensed frequency bands. Hence, architectural heterogeneity requires a dynamic, adaptable, and synchronized MAC protocol to ensure consistency in performance and reliability.

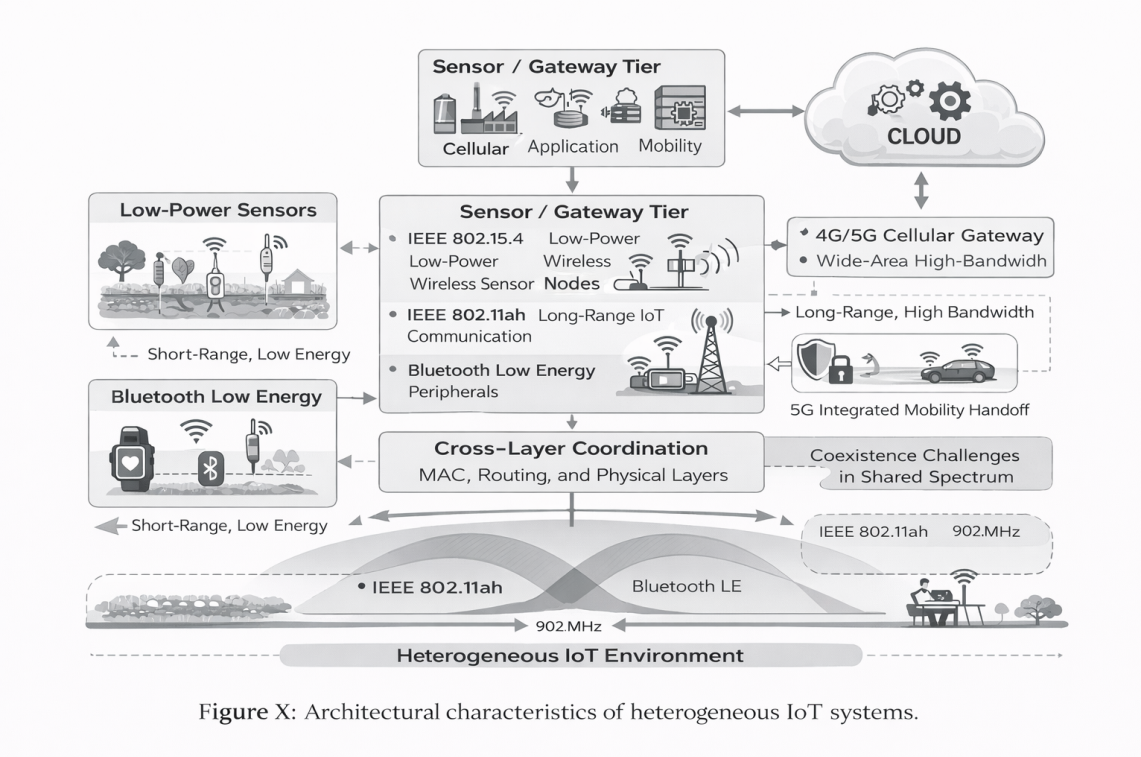

Figure X: Architectural characteristics of heterogeneous IoT systems.

Figure 1: Architectural characteristics of heterogeneous IoT systems illustrating integration of IEEE 802.15.4 low-power sensors, IEEE 802.11ah long-range IoT devices, Bluetooth Low Energy peripherals, 4G/5G cellular gateways, cross-layer coordination, and coexistence challenges in shared spectrum environments.

2.2 Performance Metrics for MAC Evaluation

The performance of the MAC protocol in heterogeneous IoT networks depends on various performance metrics, and these metrics are interdependent on each other. The energy consumption of the network is a major performance metric, taking into account the power constraints of the sensors in the network. Throughput and packet delivery ratio are major performance metrics for the reliability of the data transmission process. Delay and jitter metrics are based on the Quality of Service of the network. Fairness index depends on the efficient utilization of the channels by heterogeneous devices, and collision probability depends on the effectiveness of the contention management process. These metrics are conflicting in nature, i.e., to improve the throughput, the collision probability and energy consumption of the network might be increased.

The empirical results have shown improvements in these parameters with the help of specialized MAC optimization techniques. Hassan et al. have shown improvements in these parameters, i.e., energy consumption and retransmission overhead, with the help of deterministic TDMA-based scheduling for WBAN networks [7]. Gul et al. have shown improvements in these parameters, i.e., network lifetime, with the help of traffic-aware clustering, ensuring balanced distribution of energy consumption for cluster heads

to avoid node depletion [6]. Throughput fairness in heterogeneous traffic scenarios is improved with the help of reinforcement learning-based MAC adaptation techniques, ensuring dynamic adaptation of contention parameters to respond to channel state changes [13, 9].

Besides these primary measures, the importance of scalability and adaptability as evaluation criteria of heterogeneous IoT environments has been increasingly acknowledged. Scalability of the MAC protocol can be defined as the ability of the protocol to ensure stable performance under increasing node density, and adaptability as the ability of the protocol to respond to changes in the environment, such as changes in the amount of traffic, mobility, and interference. Distributed reinforcement learning methods enhance the scalability of the MAC protocol by avoiding excessive control overhead [4]. In addition, the optimization of the IEEE 802.11ah protocol by means of machine learning methods shows the adaptability of the protocol under dense smart city traffic conditions [8].

Another key evaluation criterion is the trade-off between energy efficiency and delay performance. Multi-objective optimization techniques strive to reach the Pareto optimality level in the trade-off between conflicting optimization goals, such as minimizing energy efficiency and minimizing end-to-end delay. Nebti and Redjimi utilized the Salp Swarm optimization technique to find the trade-off between conflicting optimization goals in heterogeneous wireless networks [11]. These optimization techniques emphasize the importance of a comprehensive evaluation framework that considers various parameters such as energy efficiency, reliability, fairness, scalability, and adaptability to evaluate the performance of heterogeneous IoT MAC protocols.

2.3 Theoretical Foundations of Optimization

Optimization theory is a mathematical foundation for resolving the trade-off problem in heterogeneous MAC protocol design. Multi-objective optimization techniques allow the simultaneous minimization of energy consumption and delay while maintaining the performance of throughput. Nebti and Redjimi proposed the application of the Salp Swarm Optimization algorithm to find the optimal solution in the Pareto set for the trade-off between energy consumption and delay in heterogeneous wireless networks [11]. The evolutionary algorithms iteratively improve the solution to find the optimal trade-off without the need for the development of analytical models of the interactions in the network.

Besides one-layer optimization, the design principles of the heterogeneous WSNs involve the coordination of the routing, physical, and MAC layers of the wireless sensor networks, thereby improving the performance of the heterogeneous wireless sensor networks. Coordination of the routing and MAC layers reduces the number of redundant transmissions, thereby improving the efficiency of the utilization of the energy of the wire-

less sensor networks. The heterogeneous wireless sensor networks are characterized by variations in link quality, mobility, and traffic, making the proposed approaches effective in improving the performance of the heterogeneous wireless sensor networks.

3 Core Concepts and Approaches

3.1 TDMA-Based Deterministic Scheduling

Time Division Multiple Access (TDMA), in which the time of communication is divided into fixed and non-overlapping slots and allocated to different nodes, avoids the issue of collision altogether. In heterogeneous wireless IoT networks, the concept of deterministic scheduling would be highly beneficial, especially for applications with specific delay and reliability constraints. Hassan et al. proposed the implementation of priority-aware superframe structures in Wireless Body Area Networks, where time slots are allocated based on the urgency of the traffic and the priority of the nodes [7]. The deterministic scheduling reduces the overhead of retransmission and idle listening by avoiding simultaneous transmissions altogether. It would be highly beneficial for the management of medium contention in environments where the transmission of safety-critical information needs to be carried out without delay.

However, although TDMA-based MAC protocols provide reliability and stability, they could impose synchronization overhead and lack flexibility, particularly in highly dynamic traffic environments. Slot allocation should take into account diverse traffic, node capabilities, and channel conditions. In a highly heterogeneous and large-scale network, there could be synchronization overhead for the entire network. Nevertheless, for latency-critical domains such as healthcare and industrial automation, deterministic scheduling remains a fundamental approach for optimizing the MAC protocol, given its capability to support guaranteed performance metrics.

3.2 Reinforcement Learning-Based MAC Protocols

Reinforcement learning (RL) enables the introduction of adaptive intelligence in the design of MAC protocols, where the wireless nodes learn the optimal transmission strategies by interacting with the wireless environment. Contrary to traditional contention-based MAC protocols, which are static in nature, RL-based MAC protocols are adaptive in nature, where the MAC protocol parameters, i.e., contention window, transmission probability, and duty cycle, are updated depending on the wireless environment. Yu et al. used deep reinforcement learning for the dynamic optimization of CSMA protocol parameters for improving fairness and throughput in heterogeneous wireless networks [13]. The channel access problem is formulated as a Markov decision process, and the wireless

nodes learn the optimal policies that maximize the long-term reward.

Another approach to improve scalability in heterogeneous IoT networks is distributed reinforcement learning. Dutta and Biswas have proposed a distributed RL-based MAC protocol for heterogeneous IoT networks, showing its effectiveness in terms of reduced retransmissions and adaptability without the need for a centralized approach [4]. RL-BMAC is a protocol that improves the previous idea by incorporating adaptive duty cycling to minimize idle listening and improve network lifetime [9]. Even though these RL-based MAC protocols are highly adaptable, they should address issues related to convergence stability, computational costs, and training latency. However, it appears that reinforcement learning is the most promising approach for dynamic MAC protocol optimization.

3.3 Machine Learning for Collision Mitigation

Machine learning algorithms can be leveraged to offer predictive capabilities, which can be integrated into the MAC parameter optimization process in dense and heterogeneous IoT networks. The machine learning models can be trained to predict traffic congestion in the network, thus adjusting the contention parameters accordingly. Ibrahim and Hafez proposed the application of adaptive machine learning models in IEEE 802.11ah-based smart city networks, which showed a significant reduction in the collision probability in dense traffic conditions [8].

Survey research also highlights the increasing trend of using machine learning in MAC protocol optimization for WSNs [2]. The learning-based mechanisms can recognize patterns, interference, and trends of node behaviors, which might not be easily captured by deterministic algorithms. However, there are challenges associated with machine learning, which include dataset availability, generalization, and computational complexity. Ensuring robustness in different heterogeneous environments is a major research issue, especially when there is a significant variation of traffic patterns over time.

3.4 Cluster-Based and Cross-Layer Strategies

Cluster-based routing strategies are beneficial for scalability and energy efficiency. In heterogeneous IoT systems, the aggregation of data is done at the cluster head level. Gul et al. proposed a traffic-aware cluster-based routing protocol that improves the network's lifetime by selecting the cluster head based on the residual energy and traffic intensity [6]. In the routing protocol, the clustering approach prevents the formation of hotspots and enhances the network's lifespan.

Cluster-based routing strategies are beneficial for scalability and energy efficiency. In heterogeneous IoT systems, the aggregation of data is done at the cluster head level. Gul et al. proposed a traffic-aware cluster-based routing protocol that improves the network's

lifetime by selecting the cluster head based on the residual energy and traffic intensity [6]. In the routing protocol, the clustering approach prevents the formation of hotspots and enhances the network's lifespan.

4 Comparative Discussion

4.1 Energy Efficiency Comparison

The optimization of energy is still a primary goal in the design of heterogeneous wireless IoT networks due to the limited energy supply of the sensors and the embedded devices. The deterministic TDMA-based MAC protocols reduce the number of collisions requiring retransmissions through the allocation of specific time slots for transmission. This method of scheduling has been shown to be effective in minimizing idle listening and the wastage of energy in contention-based access [7]. In the design of latency-critical networks, such as Wireless Body Area Networks, the scheduling of the network using the TDMA method has the effect of stabilizing the energy consumption of the network.

Reinforcement learning-based protocols also improve energy efficiency by optimizing the duty cycles and contention window sizes based on environmental feedback. RL-BMAC shows the effectiveness of sleep-wake scheduling in reducing idle listening and improving the network's lifetime [9]. Evolutionary optimization approaches, like the Salp Swarm optimization method, optimize the tradeoff between energy and delay by finding the Pareto optimal solution in heterogeneous environments [11]. Finally, the proposed intelligent routing, like the traffic-aware cluster-based routing, avoids the depletion of cluster heads by distributing the network's communication burden, thus improving the network's lifetime [6]. These intelligent routing and scheduling approaches confirm the effectiveness of the proposed energy efficiency improvements.

Table 1: Comprehensive Comparative Analysis of Energy Optimization Techniques in Heterogeneous IoT MAC Protocols

Protocol	**Core Technique**	**Optimization Objective**	**Performance Impact**
TDMA-MAC [7]	Deterministic Superframe Slot Allocation	Collision elimination, bounded latency, reliability enhancement	Reduced retransmission energy, improved packet delivery ratio, predictable delay performance in WBAN

Protocol	Core Technique	Optimization Objective	Performance Impact
RL-BMAC [9]	Reinforcement Learning-based Adaptive Duty Cycling	Dynamic energy optimization under heterogeneous traffic	Extended network lifetime, reduced idle listening, improved throughput stability
Salp Swarm Optimization [11]	Multi-objective Evolutionary Optimization	Energy–Delay Pareto trade-off balancing	Balanced residual energy distribution and improved delay-energy equilibrium
Cluster-based Routing [6]	Traffic-aware Hierarchical Clustering	Load balancing and hotspot prevention	Improved lifetime stability and scalable communication
X-MAC [3]	Short Preamble Sampling Mechanism	Idle listening reduction in low traffic networks	Lower energy consumption compared to B-MAC
RL-based MAC [5]	Q-learning Contention Window Adaptation	Collision minimization and fairness improvement	Reduced collision probability and improved fairness index

4.2 Throughput and Fairness Evaluation

Throughput and fairness are key performance indicators in the heterogeneous IoT environment, which is characterized by asymmetric traffic loads and device capabilities. Deep reinforcement learning for the optimization of CSMA protocols has been shown to enhance fairness by adapting the contention window sizes in response to the channel states [13]. The channel capture effect is mitigated by the adaptive mechanism, and fairness in bandwidth allocation for heterogeneous devices is ensured. Reinforcement learning facilitates autonomous decision-making for the IoT devices, enabling them to converge towards an optimal solution.

The IEEE 802.11ah adaptation, based on machine learning techniques, improves throughput performance even further by minimizing the probability of collision, particularly in a highly dense environment such as a smart city [8]. In this case, predictive traffic modeling optimizes the parameters of the MAC protocol to minimize contention, thus ensuring minimal loss due to collision. Deterministic TDMA scheduling ensures a guaranteed delay and predictable throughput performance [7]. Nevertheless, although TDMA ensures fairness in terms of resource allocation, it might not be the most efficient in terms of resource utilization, particularly in cases of low traffic or bursty traffic.

4.3 Scalability and Adaptability

Scalability is an essential feature that is required to cater to the needs of large-scale heterogeneous IoT systems that can contain thousands or even millions of interconnected devices spread across large geographical areas. As the scale of the network increases, the use of centralized coordination techniques becomes inefficient due to the overhead of signal transmission and the possibility of accumulated latency and single-point failures. The limitation of the centralized coordination mechanism has been overcome with the use of distributed reinforcement learning frameworks that allow each node to individually modify its MAC strategies based on the information it receives locally. Dutta and Biswas showed that the use of distributed learning mechanisms allows low-complexity IoT transceivers to modify transmission probabilities individually [4].

By reducing control signals and enabling decentralized adaptation, such frameworks greatly enhance scalability while maintaining efficient utilization of channels. Such distributed models are well-suited for industrial automation, smart cities, and urban IoT scenarios, where centralized scheduling can cause latency bottlenecks and communication congestion.

Besides distributed learning, cross-layer optimization also increases adaptability through the integration of information from the routing, MAC, and physical layers. For example, the integration of routing and MAC layers allows the network to dynamically respond to changes in the network topology, varying traffic conditions, and mobile nodes. Moreover, the integration of routing information, such as link quality and residual energy, into the MAC scheduling process can help reduce redundant transmissions, hence increasing the energy efficiency of the network. Evolutionary multi-objective optimization algorithms increase the adaptability of the network through the continuous adaptation of the routing and MAC parameters to maintain the optimal energy-delay trade-off in heterogeneous networks [11].

These algorithms aim at finding the Pareto optimal solutions under different network conditions, thus providing balanced performance with respect to different objectives. Although the inclusion of adaptivity and cross-layer approaches adds complexity in the implementation, it offers greater resilience, fault tolerance, and scalability with respect to the traditional MAC configurations. Thus, the concept of distributed intelligence and the inclusion of cross-layer approaches are of prime importance with respect to the development of sustainable scalability in the heterogeneous IoT networks of the future.

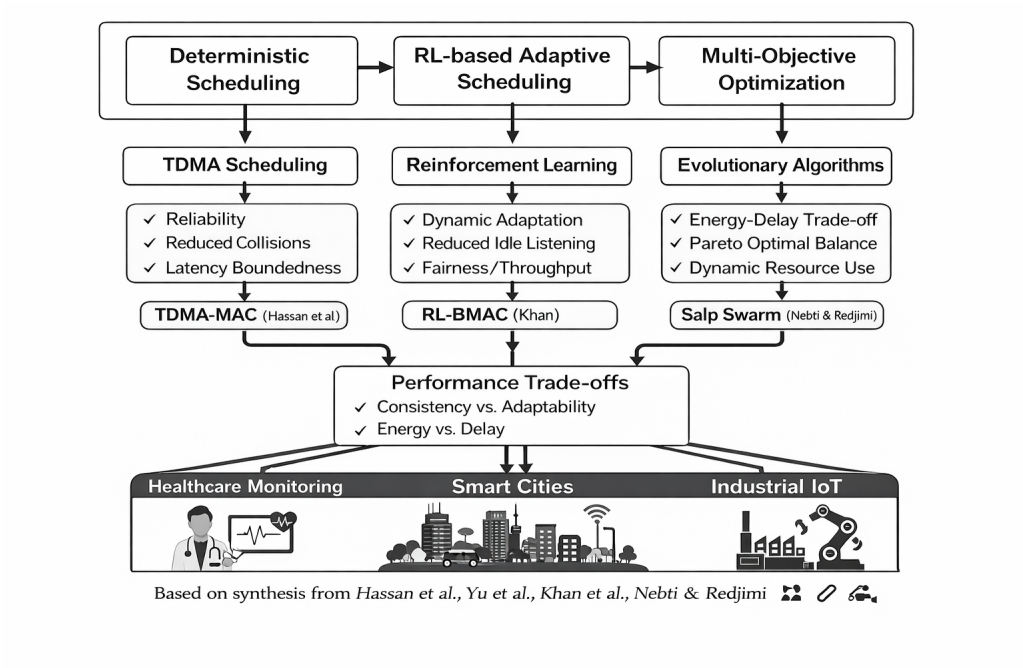

Figure 2: Comparative framework illustrating deterministic scheduling, reinforcement learning-based adaptation, and multi-objective optimization in heterogeneous IoT networks synthesized from [7, 13, 9, 11].

5 Practical Insights and Use Cases

5.1 Healthcare and Body Area Networks

Wireless Body Area Networks (WBANs) represent one of the most latency-sensitive and reliability-critical applications of heterogeneous wireless IoT systems. Medical sensors continuously monitor physiological parameters such as heart rate, blood glucose levels, and blood pressure, transmitting data to gateway devices for real-time analysis. In such environments, packet loss or transmission delay can directly affect patient safety. TDMA-based MAC scheduling provides deterministic channel access, ensuring collision-free communication and bounded latency for life-critical medical data [7]. The structured superframe design reduces idle listening and retransmission overhead, thereby enhancing both reliability and energy efficiency in constrained body-worn devices.

Furthermore, heterogeneous medical traffic often includes both emergency and non-critical data streams. Prioritized MAC mechanisms enable differentiated service levels, ensuring that emergency packets receive immediate transmission opportunities [12]. This traffic-aware scheduling enhances Quality of Service by minimizing latency for critical events while maintaining overall network stability. Hybrid architectures combining deterministic slot allocation with adaptive traffic prioritization are particularly well-suited for next-generation WBAN systems, where device heterogeneity and dynamic medical workloads are increasingly common.

5.2 Smart Cities and Dense IoT Deployments

Smart city environments are characterized by dense deployments of heterogeneous IoT devices including environmental sensors, surveillance systems, smart meters, and vehicular communication nodes. These devices generate highly variable and heterogeneous traffic patterns, often competing for limited wireless spectrum resources. Adaptive IEEE 802.11ah MAC optimization techniques significantly reduce collision probability and improve throughput in such dense deployments by dynamically adjusting contention parameters [8]. Machine learning-based traffic prediction models enhance performance by anticipating congestion and proactively modifying channel access strategies.

Reinforcement learning-based CSMA further improves fairness and channel utilization in heterogeneous urban scenarios. Deep reinforcement learning enables devices to autonomously adjust backoff intervals based on observed channel states, mitigating channel capture effects and improving throughput balance [13]. Such intelligent adaptation is particularly beneficial in dynamic urban environments where traffic intensity fluctuates due to mobility, peak usage periods, and environmental interference. Integrating learning-based MAC optimization with edge computing infrastructure can further enhance scalability and responsiveness in smart city ecosystems.

5.3 Industrial and Agricultural IoT Systems

Industrial IoT systems require scalable and reliable communication frameworks to support automation, predictive maintenance, and process monitoring. These deployments often involve heterogeneous devices operating under strict latency and reliability constraints. Distributed reinforcement learning-based MAC protocols enhance scalability by enabling decentralized decision-making among nodes with low-complexity transceivers [4]. Such distributed adaptation reduces reliance on centralized coordination while maintaining efficient medium access control under varying traffic conditions.

In agricultural IoT systems, sensor networks monitor soil moisture, temperature, humidity, and crop health over wide geographic areas. Energy efficiency and network lifetime are critical considerations due to limited accessibility and maintenance constraints. Traffic-aware cluster-based routing mechanisms improve lifetime by distributing communication load across cluster heads and reducing redundant transmissions [6]. Multi-objective optimization techniques further enhance backbone routing performance by balancing energy consumption and delay in heterogeneous wireless networks [11]. These strategies collectively enable sustainable and scalable communication infrastructures for large-scale agricultural monitoring.

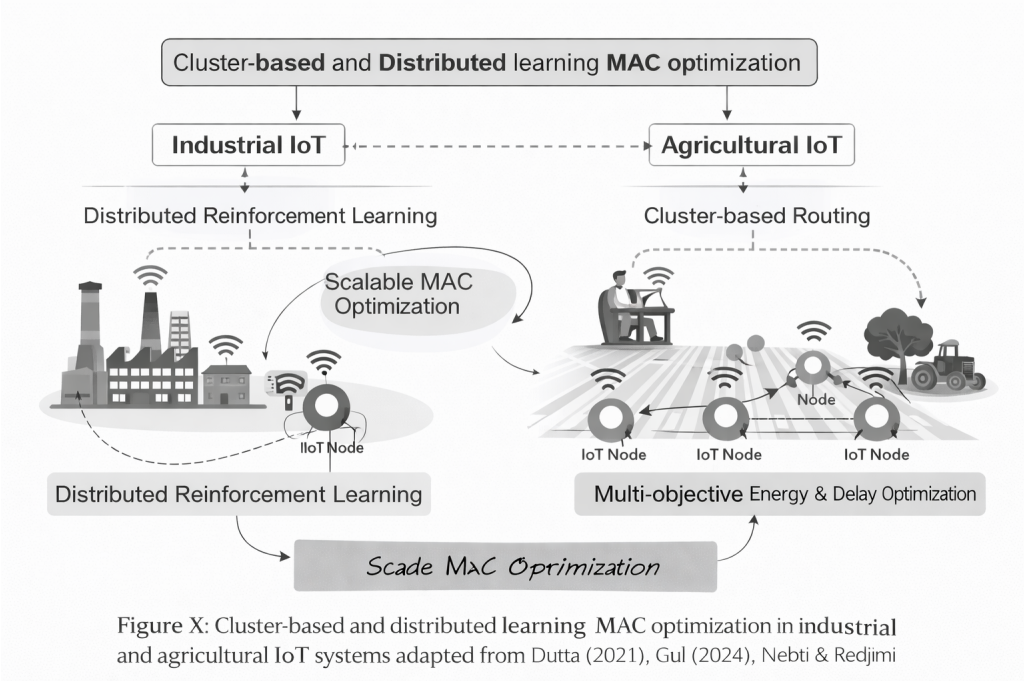

Figure 3: Cluster-based and distributed learning MAC optimization in industrial and agricultural IoT systems adapted from [4, 6, 11].

6 Challenges and Open Issues

6.1 Algorithmic Complexity and Convergence

The convergence time and computational resources are inherently involved in the implementation of reinforcement learning-based MAC protocols, and the learning algorithms have to explore the state-action space before converging to near-optimal policies. In the case of energy-constrained IoT devices, the computational resources involved in updating Q-values, neural network inference, or policy optimization could lead to increased energy consumption. In the case of the RL-BMAC protocol, the network lifetime is improved by incorporating adaptive duty cycling, and its implementation involves a tradeoff between the accuracy of the learning algorithm and resource utilization [9]. These issues are more critical for heterogeneous networks with multiple devices adapting and learning simultaneously.

Similarly, multi-objective evolutionary optimization algorithms used in the optimization of MAC and routing protocols may suffer from convergence delays. Other optimization techniques, like the Salp Swarm optimization method, aim to achieve the Pareto front trade-off between the delay and energy consumption. However, evolutionary-based optimization techniques may take several iterations before achieving convergence stability [11]. This may be a challenge in dynamic IoT environments where the traffic conditions change frequently. Moreover, evolutionary-based optimization techniques may also suffer from memory and computational overheads, which may not be suitable for ultra-low-

power devices.

6.2 Synchronization and Scalability

For instance, deterministic scheduling techniques such as TDMA ensure robust and safe communication without the threat of collisions. However, the synchronization between the participating devices in the network also becomes an added control overhead, especially in large-scale networks. Hassan et al. demonstrated the applicability and effectiveness of the TDMA scheduling protocol in the context of wireless body area networks. However, the applicability and effectiveness of the protocol in the context of large-scale heterogeneous IoT networks are questionable, considering the control overhead in maintaining the synchronization between the devices in the network [7]..

Scalability becomes more complex as heterogeneous nodes with different capabilities are present in the same communication environment. The cross-layer optimization strategies are also proposed to enhance the performance of the heterogeneous network with the integration of routing and MAC layers. The complexity of the protocol is also increased with the integration of these strategies. In the heterogeneous 5G-integrated IoT network, the mobility-aware handoff also becomes more complex while performing the MAC coordination [10]. Scalability with performance guarantees is another open research issue.

6.3 Standardization and Benchmarking

Although notable improvements have been achieved in the field of MAC protocol optimization, there is a lack of standardized benchmarking for heterogeneous IoT networks. In many cases, the performance improvement of the proposed techniques is validated in a controlled simulation environment with a certain traffic pattern, node density, or channel model. These frameworks for evaluating performance improvements might not accurately represent the heterogeneity of real-world networks. In addition, multi-objective optimization techniques have shown improvements in the energy-delay trade-off; however, a standardized framework for benchmarking is essential to compare the performance improvement of different techniques [11].

Another underdeveloped field in this regard is statistical validation of performance enhancements. Reinforcement learning-based MAC protocols and machine learning-based optimizations of IEEE 802.11ah protocols often do not perform a thorough analysis of confidence intervals or hypothesis tests for assessing their enhancements [9, 8]. In order to promote generalization, it is important that future benchmarking protocols include multi-scenario testing, sensitivity analysis, and datasets with diverse traffic and interference patterns. The development of open experimental testbeds will be vital for advancing MAC optimization research.

7 Future Directions

7.1 Hybrid Adaptive Architectures

For future MAC protocol research, there is a need for hybrid MAC protocols with a combination of determinism and adaptability for heterogeneous wireless IoT networks. Although TDMA-based MAC protocols have bounded end-to-end delay guarantees and performance guarantees, especially in safety-critical applications like WBANs [7], deterministic MAC protocols do not offer adaptability under bursty or dynamic traffic conditions. On the other hand, reinforcement learning-based MAC protocols have good adaptability in heterogeneous contention environments, but there is a convergence delay and computational complexity [13, 9].

This could be achieved through the development of a hybrid system that can allocate deterministic time slots for latency-critical or high-priority traffic while simultaneously employing adaptive contention-based techniques for best-effort communications. In the case of WBAN systems, for example, emergency medical information could be allocated specific time slots using a TDMA system, while reinforcement learning-based contention window management could be used for less critical background traffic [12].

Besides, the distributed reinforcement learning approaches facilitate the adaptation process, thus making the system scalable, especially for the IoT environment [4]. In addition, the hybrid system may be able to achieve the optimization of multiple objectives, including minimizing delay, energy, and collision probability. Evolutionary optimization methods, like the Salp Swarm optimization method, have been shown to be effective in optimizing multiple performance measures, especially in heterogeneous wireless systems [11]. These optimization methods may be used in hybrid MAC systems, thus ensuring parameter tuning without compromising the deterministic performance.

Another important aspect of hybrid design is cross-layer coordination. Coordination of routing, mobility management, and MAC scheduling can improve system performance considerably in heterogeneous 5G-integrated IoT networks [10, 1]. Hence, in future hybrid MAC designs, there is a need for cross-layer information exchange for overall optimization of different layers of protocols..

7.2 Edge Intelligence and Energy Harvesting

The increasing integration of edge computing in IoT architectures presents new opportunities for MAC protocol optimization. Learning-based MAC schemes often require computationally intensive training and decision-making processes, which may exceed the capabilities of resource-constrained sensor nodes. Offloading reinforcement learning model updates or traffic prediction computations to edge servers can significantly reduce energy overhead at end devices [9]. Edge intelligence enables near real-time optimization while

preserving scalability in heterogeneous environments.

Machine learning-based collision mitigation strategies, such as adaptive IEEE 802.11ah parameter tuning, can also benefit from edge-assisted training pipelines [8]. Edge nodes can aggregate traffic statistics from multiple heterogeneous devices, train predictive models, and distribute optimized MAC configurations to network nodes. This centralized learning with distributed execution model enhances both accuracy and energy efficiency.

In parallel, energy harvesting technologies are increasingly integrated into IoT deployments, enabling nodes to replenish energy from environmental sources such as solar, thermal, or kinetic energy. However, energy harvesting introduces variability in available power, necessitating dynamic MAC adaptation strategies. Adaptive duty cycling mechanisms, as demonstrated in RL-BMAC, can be extended to consider harvested energy profiles [9]. MAC protocols may incorporate predictive energy availability models to adjust transmission schedules and contention parameters accordingly.

Cluster-based routing combined with energy-aware scheduling further enhances sustainability in energy harvesting-enabled networks. Traffic-aware clustering strategies distribute communication load to prevent premature depletion of critical nodes [6]. Integrating energy harvesting awareness into clustering decisions represents a promising direction for long-term heterogeneous IoT sustainability.

7.3 Statistical Validation and Multi-Scenario Testing

Although substantial improvements in performance have been achieved by a number of MAC optimization techniques, the standard statistical validation of these improvements remains underdeveloped in the context of heterogeneous wireless IoT networks. In fact, many of the existing works have been validated via simulation under single scenario configurations, and thus their results cannot be generalized for a wide variety of heterogeneous network deployments. Hence, it is essential to incorporate statistical validation techniques, such as statistical hypothesis testing, confidence intervals, and variance analysis, to support the performance improvements achieved in terms of energy efficiency, delay reduction, and throughput enhancement. In fact, multi-objective optimization frameworks have already been proposed to address the trade-offs between energy efficiency and delay reduction, ensuring pareto optimality for these two important metrics [11]. In addition, the performance of reinforcement learning-based MAC protocols should be validated in terms of convergence stability, sensitivity to initial parameters, and long-term robustness to stochastic changes in traffic characteristics [9].

Similarly, the IEEE 802.11ah optimization strategies based on machine learning models should be cross-checked using heterogeneous smart city traffic data to test the generalization and adaptability of the models beyond the experimental setup [8]. Moreover, the benchmarking of deterministic scheduling, reinforcement learning-based adap-

tation, clustering-based optimization, and cross-layer optimization strategies should be conducted using the same simulation setup to facilitate comparative analysis.

Experimental validation using real-world testbeds would significantly enhance credibility beyond simulation-based evaluation. Integration with 5G heterogeneous infrastructure and mobility-aware handoff scenarios should be included in validation pipelines [10]. Establishing open benchmarking datasets and reproducible simulation models will accelerate progress and ensure transparency in MAC optimization research.

In summary, the future of MAC protocol research should address performance metrics in a manner that is not only statistically validated, but also involves multiple scenarios and experimental reproducibility. Hybrid multi-objective architectures, along with validation methodologies, offer the most promising approach for the development of heterogeneous wireless IoT networks.

8 Conclusion

The optimization of the Medium Access Control protocol is an intricate research problem with multidimensional aspects for heterogeneous wireless IoT networks. The presence of different energy capacities, transmission ranges, computational capabilities, and Quality of Service demands of the devices alters the conventional MAC protocol design paradigm. In addition to this, the heterogeneous wireless IoT networks are different from the homogeneous wireless sensor networks in terms of the ability to fulfill the reliability, scalability, energy efficiency, fairness, and latency demands simultaneously.

Deterministic scheduling algorithms, such as Time Division Multiple Access schemes, offer guaranteed access and predictable access delays. Hassan et al. proved that structured superframe-based TDMA reduces packet collision probability and idle listening for Wireless Body Area Networks [7]. Other prioritization schemes for diverse body area network traffic improve latency performance for emergency transmissions [12]. Deterministic scheduling algorithms are highly reliable but are limited by increased synchronization overhead and lower adaptability to highly dynamic traffic conditions.

In contrast, the role of adaptive intelligence through reinforcement learning has been recognized as an effective means to optimize MAC parameters. Deep reinforcement learning has been applied to the CSMA protocol to improve the fairness of throughput and reduce the effect of capture in heterogeneous contention environments [13]. Distributed reinforcement learning has also been proposed to improve scalability for low-complexity IoT transceivers [4]. The RL-BMAC protocol has also shown that the use of adaptive duty cycling can significantly improve the network lifetime [9]. Although the use of learning-based solutions has been recognized as an effective means to improve adaptability, it also introduces complexity and convergence issues that require careful consideration.

Machine learning-based MAC optimization also increases the effectiveness of MAC op-

timization in terms of collision mitigation in dense network deployments. The application of learning models for adaptive parameter tuning of IEEE 802.11ah protocols minimizes the probability of collisions in smart city IoT networks, which often experience diverse traffic patterns [8]. Survey studies also support the fact that machine learning is becoming a key component of MAC optimization in WSNs [2]. However, training overhead is still a research issue.

Besides single-layer optimization, clustering and cross-layering also offer further enhancements in the performance of the system. Traffic-aware cluster-based routing is used to optimize the communication load and increase the network's lifetime in IoT-assisted WSNs [6]. Multi-objective Salp Swarm optimization is also used to achieve a balance between the energy consumption and delay in heterogeneous wireless networks [11]. Cross-layering also increases the efficiency of the system by minimizing the energy consumption through the synchronization of the routing and MAC layers [1]. Besides, in heterogeneous 5G-integrated IoT networks, the mobility management also has a considerable effect on the MAC efficiency in the handover process [10].

Future MAC protocol designs should integrate TDMA-based reliability for critical traffic with reinforcement learning-driven adaptability for best-effort communication. Multi-objective optimization techniques must be employed to balance energy efficiency, delay, and fairness simultaneously. Moreover, rigorous statistical validation frameworks are necessary to evaluate performance improvements across heterogeneous deployment scenarios and ensure robustness beyond simulation-based evaluation [11, 9]. Standardized benchmarking methodologies and real-world experimental validation will further strengthen the credibility of MAC optimization research.

In conclusion, in order to tackle the heterogeneity issue in wireless IoT networks, there is a need for a balanced blend of determinism and adaptability. The best solution for developing scalable, energy-efficient, and reliable heterogeneous IoT communication networks is through hybrid multi-objective MAC protocols with statistical validation.

References

[1] Sandhya Onkar Ahire. Cross-layer design approaches for enhanced energy efficiency in heterogeneous wireless sensor networks (wsns).

[2] Noor Zuriatunadhirah binti Zubir, Aizat Faiz Ramli, and Hafiz Basarudin. Optimization of wireless sensor networks mac protocols using machine learning; a survey. In *2017 International Conference on Engineering Technology and Technopreneurship (ICE2T)*, pages 1–5. IEEE, 2017.

[3] Michael Buettner, Gary V Yee, Eric Anderson, and Richard Han. X-mac: a short preamble mac protocol for duty-cycled wireless sensor networks. In *Proceedings of the*

4th international conference on Embedded networked sensor systems, pages 307–320. ACM, 2006.

[4] Hrishikesh Dutta and Subir Biswas. Medium access using distributed reinforcement learning for iots with low-complexity wireless transceivers. In *2021 IEEE 7th World Forum on Internet of Things (WF-IoT)*, pages 356–361. IEEE, 2021.

[5] Ana Galindo-Serrano and Linda Giupponi. Reinforcement learning-based mac protocol for wireless sensor networks. *IEEE Communications Letters*, 23(3):516–519, 2019.

[6] Hina Gul, Sana Ullah, Kim Ki-Il, and Ali Farman. A traffic-aware and cluster-based energy efficient routing protocol for iot-assisted wsns. *Computers, Materials, & Continua*, 80(2):1831, 2024.

[7] Wan Haszerila Wan Hassan, Darmawaty Mohd Ali, J Mohd Sultan, and Murizah Kassim. Medium access control protocol based on time division multiple access scheme for wireless body area network. *Int. J. Electr. Comput. Eng.*, 14(3):2762, 2024.

[8] Amin Ibrahim and Ahmed Hafez. Adaptive ieee 802.11 ah mac protocol for optimization collision probability in iot smart city data traffic based machine learning models. 2023.

[9] Owais Khan, Sana Ullah, Muzammil Khan, and Han-Chieh Chao. Rl-bmac: An rl-based mac protocol for performance optimization in wireless sensor networks. *Information*, 16(5):369, 2025.

[10] Kiran Mannem, Pasumarthy Nageswara Rao, and S Chandra Mohan Reddy. Power optimized intelligent handoff mechanism for 5g-heterogeneous network. *Multimedia Tools and Applications*, 83(19):56697–56718, 2024.

[11] Salima Nebti and Mohammed Redjimi. Multi objective salp swarm based energy efficient routing protocol for heterogeneous wireless networks. *International journal of Computer Networks & Communications*, 15:73–92, 2023.

[12] Shama Siddiqui and Indrakshi Dey. Optimizing next generation wireless ban with prioritized access for heterogeneous traffic. In *2025 International Conference on Computer, Information and Telecommunication Systems (CITS)*, pages 1–5. IEEE, 2025.

[13] Yiding Yu, Soung Chang Liew, and Taotao Wang. Carrier-sense multiple access for heterogeneous wireless networks using deep reinforcement learning. In *2019 IEEE*

Wireless Communications and Networking Conference Workshop (WCNCW), pages 1–7. IEEE, 2019.